

Anurag Singh

AI/ML Engineer | Natural Language Processing | Deep Learning | Python
Transformer fine-tuning | RAG pipelines | Applied NLP research | Machine Learning
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EDUCATION

Shri Mata Vaishno Devi University — B.Tech, Computer Science and Engineering Aug 2023 – May 2027
• CGPA: 7.5/10 (through Semester 6) | Coursework: Deep Learning, Natural Language Processing, Machine Learning, Data Structures & Algorithms

RESEARCH & PUBLICATIONS

RoBERTa-Driven Transformer Framework for Context-Aware Sentiment Classification of Social Media Text

First Author | Accepted — NLPiR 2026 (Nara, Japan)

Fine-tuned RoBERTa for 3-class sentiment classification on 18,170 YouTube comments achieving 96% accuracy, 0.96 weighted F1, 0.95 macro-F1. Resolved class imbalance (62.6% vs 12.9%) via weighted cross-entropy, label smoothing, AdamW with LR scheduling on a stratified 70/15/15 split using PyTorch and Hugging Face Transformers.

RAG-RERoBERTa: Explainable Multilingual Biomedical Question Answering

Undergraduate Research | In Progress

Designed a Recursive Embedding Refinement mechanism targeting +4–8% accuracy over BioBERT/PubMedBERT baselines with 5-stage curriculum training across 3 biomedical QA datasets (290K+ samples). Integrated GAT evidence aggregation, SHAP/Integrated Gradients explainability, and multilingual English–Hindi inference with neural TTS.

EXPERIENCE

Software Development Intern — Eduspheria | Remote

Apr 2026 – Present

- Integrated AI features into Saksham, a 97K-LOC / 60+ page React 18 platform: DALL·E 3 poster generation, AI question-paper generation, and a WebSocket-based conversational assistant.
- Implemented role-based access control across 6 user roles and consumed 30+ REST endpoints (Django REST Framework) with token authentication.
- Designed 5+ gamified learning exercises and integrated 100+ PhET simulations via a postMessage architecture with TTS narration (ElevenLabs), used by students in Grades 5–12.

Freelance Software & AI Engineer — Independent | Remote

Dec 2023 – Present

- Delivered AI/ML, NLP, and RAG-pipeline projects for clients — LLM APIs, embeddings, vector databases, and end-to-end machine learning workflows.
- Built backend and web applications using Python, Django, Django REST Framework, Node.js, REST APIs, and SQL/PostgreSQL.
- Conducted end-to-end ML work: dataset preparation, preprocessing, model implementation, evaluation, and technical documentation for client deliverables.

SELECTED PROJECTS

RoBERTa Sentiment Classification | Python | PyTorch | Hugging Face | RoBERTa

2026

- Fine-tuned RoBERTa transformer (PyTorch, Hugging Face) for NLP sentiment classification on 18,170 social media comments, achieving 96% accuracy and 0.95 macro-F1 — accepted, NLPiR 2026.

NL2SQL: Text-to-SQL Model | Python | T5 | QLoRA | WikiSQL

2026

- Fine-tuned T5-small for natural-language-to-SQL generation on the WikiSQL dataset (80K+ query pairs), applying QLoRA to fit training within a 4GB VRAM budget.

Pragya: Local RAG Research Assistant | Llama3 | ChromaDB | MiniLM | Streamlit

2026

- Built a fully local, zero-cloud RAG system with Llama3 and MiniLM embeddings to make complex research papers accessible in both technical and lay-audience formats.
- Implemented section-aware chunking with PyMuPDF + ChromaDB, improving retrieval relevance over naive chunking; mean query latency under 400ms on CPU hardware.

TECHNICAL SKILLS

Programming:	Python, Java, C, C++, SQL, JavaScript, Data Structures & Algorithms
ML / DL:	Machine Learning, Deep Learning, MLOps, PyTorch, TensorFlow, Scikit-Learn, Hugging Face Transformers, Pandas, NumPy, MLflow, Optuna, TensorBoard, torch-geometric (GAT)
NLP:	RoBERTa, T5, Transformer Fine-Tuning, Attention Mechanisms, Tokenization, Text Preprocessing, Class-Imbalance Handling, Explainable AI (SHAP, Captum), Multilingual NLP, Text-to-Speech
LLM / GenAI:	Generative AI, LLM APIs (OpenAI, DeepSeek), LLM Fine-Tuning, Prompt Engineering, RAG Pipelines, Vector Databases (ChromaDB), Embeddings (MiniLM), AI Agents
Systems & Infra:	Linux, CUDA, Docker, Git, FastAPI, AWS

ACHIEVEMENTS & LEADERSHIP

- Smart India Hackathon 2025 Finalist** — Among 1,360 finalist teams selected from 68,766 nationwide. Built KrishiRakshak, an AI platform for early wheat disease detection fusing satellite imagery, drone surveillance, and farmer-uploaded leaf images to generate real-time disease risk maps and predictive advisories (React, Node.js, Python, PyTorch, TensorFlow, MongoDB).
- AI Club Co-Lead** — Directing AI initiatives and technical workshops on Deep Learning and Computer Vision for 100+ students at SMVDU.